

8. [Maximum mark: 10]

Let $z = 1 + \cos 2\theta + i \sin 2\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

(a) Show that

(i) $\arg z = \theta$;

(ii) $|z| = 2 \cos \theta$.

[7]

(b) Hence or otherwise, find the value of θ such that $\arg(z^2) = |z^3|$.

[3]

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